
BRAIN DRAIN DETECTOR: MULTIMODAL COGNITIVE OVERLOAD DETECTION IN CONVERSATION

Theodora Pavlidou
theod.pavlidou@gmail.com

Marilena Papasideri
marilena.papasideri@gmail.com

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[Source code](#)

ABSTRACT

Cognitive overload during high-stakes conversation impairs decision-making and communication. This report presents "Brain Drain Detector," a multimodal deep learning system designed to automatically classify 5-second segments of naturalistic dyadic debates into "Safe" or "Alarm" (cognitive overload) states. Using the K-EmoCon corpus, we extract ground-truth binary labels from continuous self-reported arousal and valence ratings. Our methodology encompasses a robust end-to-end preprocessing pipeline including speaker diarization, voice activity detection, and physiological signal windowing. To handle the highly imbalanced class distribution, we employ a class-weighted loss and balanced sampling strategy evaluated under a strict Leave-One-Subject-Out (LOSO) cross-validation protocol. We systematically compare early fusion strategies (e.g., audio-anchored pooled cross-attention) against late fusion alternatives across three distinct modalities: Audio (Wav2Vec 2.0), E4 wristband data (EDA, HR, IBI), and NeuroSky EEG. Extensive ablation studies evaluate component architectures, including a 1D CNN front-end, inter-window context, and a hand-crafted feature MLP. While the best single early-fusion network achieves a 59.50% Macro-F1 score, our final results demonstrate that Late Fusion Stacking, using a logistic regression meta-learner over independent modality predictions, achieves the best overall performance with a 61.86% Macro-F1 score. This work highlights the critical trade-offs between early and late multimodal arbitration and provides a comprehensive baseline for overload detection in naturalistic dialogue.

Keywords cognitive overload · multimodal fusion · K-EmoCon · cross-attention

1 Introduction

People under cognitive overload during conversation make poorer decisions, communicate less clearly, and may burn out over time. Detecting overload early from observable signals could support monitoring in debates, meetings, or other high stakes dialogue. This report presents Brain Drain Detector, a multimodal system that classifies short segments of conversation as **Safe** or **Alarm** using audio, wristband physiology, and EEG.

We build on K EmoCon [1], a public corpus of naturalistic dyadic debates with multimodal recordings and continuous self ratings of arousal and valence every five seconds. Most work on this dataset targets emotion recognition or regression on those ratings. We repurpose it for a binary overload detection task defined from self annotations only, with a full pipeline from raw debate audio to trained fusion models.

Our main contributions are:

1. An end to end preprocessing pipeline: speaker diarization and separation, voice activity detection, physiological windowing, and quality gated augmentation rules (Section 4.3).
2. A fair training protocol for imbalanced Safe versus Alarm classification, and a fusion design that moves from an initial balanced cross attention attempt to a simpler pooled cross attention that performs better on this dataset.
3. A systematic set of experiments over early fusion variants, late fusion strategies, biosignal encoder ablations, and unimodal baselines.

4. A modality overlap analysis that relates fusion predictions to audio only and biosignal only errors.

The rest of the report is organized as follows. Section 2 defines the task and evaluation metrics. Section 3 describes the data. Section 4 covers preprocessing, including optional offline augmentation (Section 4.3). Section 5 presents the model and training objective. Section 6 reports experimental results. Section 7 concludes.

2 Problem Formulation

To establish ground-truth labels, we utilize the continuous emotion annotations provided in the K-EmoCon corpus. Following each debate, participants reviewed their recorded sessions and self-reported their perceived **arousal** (activation level) and **valence** (positivity) on a discrete 5-point Likert scale. These ratings were sampled every five seconds, establishing a fixed temporal annotation grid across the dialogue.

Each sample in our formulation corresponds to one 5-second window aligned with this grid. We derive the binary labels exclusively from the self-annotations (`P{N}.self.csv`). For each window, given the self-reported arousal A and valence V , we assign a binary target as follows:

Label	Name	Condition
0	Safe	$V > 3$ OR $A < 4$
1	Alarm	$V \leq 3$ AND $A \geq 4$

Table 1: Binary Safe/Alarm labeling from arousal and valence.

Safe covers non overloaded states, including neutral or positive valence and lower arousal. **Alarm** marks windows where the participant reports low valence together with high arousal, which we treat as cognitive overload for this project.

K EmoCon provides three parallel label streams: self, debate partner, and external observer ratings of arousal and valence every five seconds [1]. We use **self annotations** only. The dataset authors describe self reports as the firsthand account of what the participant felt during the debate, while partner and observer ratings reflect how emotion was *displayed* or perceived from outside. Those views often disagree because people regulate, mask, or misread emotion in social interaction [1].

For modeling a participant’s own overload state from their biosignals and speech during the debate, self report is the natural target. Prior work on K EmoCon that predicts arousal and valence from wearable physiology likewise treats self reported ratings as the label to match [2]. Partner and observer files are not used in this project because our task is tied to the participant’s subjective state, not to an external judgment of their expression.

After audio voice activity filtering, the dataset contains **1,492** windows from **27** participants. The class distribution is heavily skewed: roughly **83%** Safe and **17%** Alarm (about 1,252 Safe and 240 Alarm windows). A model that always predicts Safe would achieve high accuracy but would miss almost every overload event.

2.1 Evaluation metrics

All main result tables in this report use metrics derived from a **synthetic 50/50** confusion matrix. We first pool Leave One Subject Out (LOSO) test predictions across folds, build an aggregated confusion matrix on the real class split, read row normalized recalls for Safe and Alarm, and then construct a balanced matrix with 500 Safe and 500 Alarm samples before computing metrics. This gives a fair comparison between classes when reporting Macro-F1 and per class scores. Section 6 states this protocol again where results are shown.

Macro-F1 is the primary metric for model selection and comparison. For a monitoring use case we also report **Recall for the Alarm class** and **F1 for the Alarm class**. Missing an Alarm window is more costly than a false alarm, so Alarm recall is treated as a safety relevant secondary metric. We also report Accuracy, but it is dominated by the Safe majority under the real class distribution.

3 Dataset

This section describes the **raw K EmoCon corpus** as released to researchers. Section 4 explains how we turn that corpus into aligned training windows.

K EmoCon [1] contains approximately ten minute dyadic debates on social issues. Sixteen paired sessions were recorded with 32 participants. Each session includes stereo audio, E4 wristband streams, NeuroSky EEG, and

continuous emotion ratings every five seconds. The public release also includes metadata, data quality tables, and emotion annotation files from three perspectives (self, partner, and external observers).

Table 2 lists the raw sensing streams we use. At this stage audio is a single stereo file per debate (e.g. p1.p2.wav at 16 kHz), with one channel assigned to each debater. Physiological CSV files are stored per participant and aligned in time with the debate recording.

Modality	Raw signals	Device
Audio	Stereo debate recording (left/right per participant)	Debate WAV, 16 kHz
E4	EDA, HR, IBI	Empatica E4
EEG	θ , α , β band power	NeuroSky MindWave

Table 2: Raw modalities in the K EmoCon corpus used by Brain Drain Detector.

To account for signal loss due to sensor displacement or dropped packets, the corpus provides **data completeness tables**. These tables report a score in $[0, 1]$ for each participant’s sensor streams, representing the proportion of valid data captured during the session. We utilize these quality scores to inform our data augmentation strategy (Section 4.3).

After our preprocessing pipeline (Section 4), **27** of the 32 participants yield retained five second windows that pass speech overlap filters. That filtered set contains **1,492** labeled windows with the class balance stated in Section 2.

3.1 Train and test protocol

We evaluate with **Leave One Subject Out** cross validation. In each of 27 folds, one participant is held out for testing and the model trains on all remaining participants. No window from the test participant appears in training for that fold. This protocol matches the small number of subjects and avoids optimistic estimates from random splits that could leak identity across train and test sets.

4 Preprocessing Pipeline

Section 3 described the corpus as published. This section describes our four stage preprocessing pipeline that produces model ready samples. Each final sample is one five second window with (i) a mono audio waveform for one participant, (ii) a 50×6 biosignal tensor, and (iii) a Safe/Alarm label from Table 1.

4.1 Audio separation and windowing

Speaker diarization and separation. Stereo debate files mix both speakers on two channels. Before windowing, we build a **separate mono speech track** for each participant so that each training window contains only that person’s speech.

Input. Each debate is stored as a stereo file (e.g. p1.p2.wav): the left channel belongs to the first participant and the right channel to the second.

Step 1: Diarization on the mono mix. We compute $(\text{left} + \text{right})/2$ and run PyAnnote speaker diarization 3.1 [3] with two speakers fixed. The pipeline uses neural segmentation and speaker embedding models. The output is a list of time intervals with anonymous speaker labels, indicating who spoke when.

Step 2: Map speakers to participants. Diarization does not know which speaker corresponds to which debater. For each diarized speaker we sum left channel $\text{RMS} \times \text{duration}$ and right channel $\text{RMS} \times \text{duration}$ over all of that speaker’s segments. Comparing these energy ratios across the two main speakers yields a **global** assignment to the left and right participant for the entire file. This stays stable across long debate turns.

Step 3: Build keep masks. For each interval assigned to a participant, we mark the corresponding samples on that participant’s boolean mask. Masks are dilated by about 80 ms to avoid cutting word edges. We then **mute the other speaker**: on the first participant’s track, samples outside their mask are set to zero, and similarly for the second participant.

Step 4: Fill short gaps. To prevent the diarization model from fragmenting long monologues due to brief breaths, we bridge any non-speech gap shorter than three seconds if the same participant is speaking immediately before and after the gap.

Step 5: Resolve overlap. When both masks are active during crosstalk, we keep the sample on the channel with larger absolute amplitude and zero the other.

Step 6: Cache. Diarization intervals are saved per debate. Later runs can rebuild separated tracks from the cached intervals without rerunning PyAnnote.

Voice activity detection and windowing. On each separated mono track we run PyAnnote voice activity detection once to obtain a global speech timeline for that participant.

We walk the fixed **five second annotation grid** (0 to 5 s, 5 to 10 s, 10 to 15 s, and so on). A window is **retained** when its overlap with detected speech is at least three seconds **or** at least 60% of the window length. The first kept window is annotated at the five second mark and covers audio samples in $[0, 5)$ s.

Each accepted window stores the waveform, participant identifier, annotation timestamp, and measured speech overlap in seconds.

4.2 Physiological signal windowing

E4 wristband and NeuroSky EEG streams are resampled to a uniform **10 Hz** grid with linear interpolation within each five second span. The grid matches the same annotation timestamps as the audio windows from Section 4.1.

A physio window is kept only when a matching audio window exists for the same participant and time stamp. Retention is therefore driven by the speech overlap rules on audio, not by sensor quality scores. We do **not** discard biosignal windows because E4 or EEG completeness is below 1.0. Noisy segments remain in the dataset and are handled by per window normalization at training time.

For every aligned pair we apply **z score normalization per window** (zero mean, unit variance independently for each channel in that five second span). This removes slow tonic offsets inside each segment and puts all channels on a comparable scale for the downstream biosignal encoder (Section 5).

We also tested an alternative **per participant normalization**: for each debater, mean and standard deviation of every channel are computed over *all* retained debate windows, then applied to each window from that participant. Unlike per window scaling, this preserves slow drift in EDA or heart rate baseline across the conversation. Section 5.5 reports a late fusion experiment that rebuilds tensors with this scheme.

Six channels are stacked in fixed order: EDA, HR, IBI from the E4 wristband, then θ , α , β EEG band power from NeuroSky. At 10 Hz over 5 s this yields a biosignal tensor of shape 50×6 per window.

The E4 completeness tables described in Section 3 gate **offline augmentation only** (Section 4.3): synthetic sensor noise is added only when completeness equals 1.0. The default fusion model does not read these tables. A separate experiment feeds quality scores into the fusion layer itself (Section 5.4); that is not augmentation and does not alter the stored biosignal tensors.

Accepted audio windows, aligned physio tensors, and Safe/Alarm labels are then merged into unified training records: each record contains a one dimensional audio waveform, a 50×6 biosignal tensor, the participant id, the window timestamp, and the binary label.

4.3 Optional augmentation

All main classification experiments train on the non augmented windows described above. We additionally support an **offline augmentation** path for ablation: synthetic perturbations are applied once when tensors are built and stored as a separate copy of each window, then reused every epoch without recomputation.

Physiological noise is injected only when the participant’s E4 completeness score equals 1.0 for the relevant sensors. If quality is lower, the recording already contains natural noise and extra injection would distort the signal. This use of quality is **augmentation gating only**. It must not be confused with quality aware **fusion** (Section 5.4), where the same completeness scores modulate the multimodal fusion layer during training.

Audio perturbations in the offline path include time and frequency masking (SpecAugment style) and additive Gaussian noise on the waveform.

The default training setup keeps augmentation disabled. Section 6 compares this baseline against the offline augmented tensors; gains were small on Macro-F1.

5 Model Architecture

Brain Drain Detector uses **intermediate fusion**: each modality is encoded separately, combined in a fusion module, and passed to a binary classification head. All fusion variants share a **256 dimensional** embedding space before the head. Figure placement for the overview diagram is reserved for a later revision; the flow is audio encoder $\rightarrow$ biosignal encoder $\rightarrow$ fusion $\rightarrow$ two logits (Safe / Alarm).

5.1 Modality encoders

Audio. The default audio branch is pretrained **Wav2Vec 2.0** [4]. The transformer output sequence is mean pooled to a 768 dimensional vector, then projected to 256 dimensions. During training the Wav2Vec 2.0 backbone stays **frozen**. It acts as a fixed feature extractor while the biosignal encoder, fusion layer, and classification head learn from the relatively small K EmoCon subset (27 participants in our filtered set). This limits overfitting and avoids damaging pretrained speech representations.

Biosignals. Each five second window is a multivariate sequence of shape 50×6 : three E4 channels (EDA, HR, IBI) and three EEG bands (theta, alpha, beta), sampled at 10 Hz after preprocessing (Section 4). The default encoder runs one **bidirectional GRU (BiGRU)** with hidden size 128 over all six channels jointly. Forward and backward final hidden states are concatenated, giving a 256 dimensional biosignal embedding per window. We stack two GRU layers with dropout between them. BiGRU is preferred over LSTM in our ablations because it has fewer parameters and trained more stably on 27 participants. All biosignal variants below feed the same 256 dimensional fusion space unless noted.



Figure 1: Default biosignal encoder: a single BiGRU tower over all six channels produces a 256-d embedding, then pooled cross-attention with the frozen Wav2Vec 2.0 audio branch.

5.2 Biosignal encoder ablations

We kept the audio branch and fusion stack fixed and swapped only the biosignal encoder (or, for the inter-window variant, the head on fused embeddings). Quantitative results for all variants are in Section 6.2.

1D CNN front end. The baseline biosignal encoder uses BiGRU only. In a separate ablation we prepend a lightweight **1D convolutional front end** before the BiGRU. Conv1d layers (kernel size 5, 32 output channels, ReLU, dropout) scan the six channel input along time and extract **local temporal motifs** (short EDA bursts, sharp HR changes, brief EEG band shifts) before the recurrent layer aggregates the full five second context. The motivation is that some overload cues may appear as short transients within a window rather than as a single global pattern captured only by final GRU states. This stack is **not part of the baseline model**; it underperformed the plain BiGRU encoder on 27 participants.

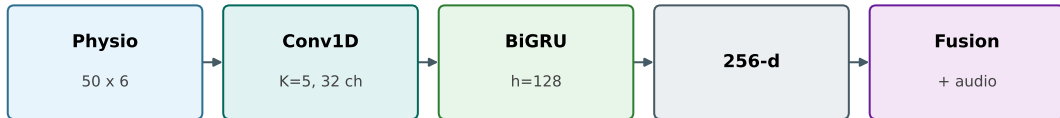


Figure 2: 1D CNN front-end ablation: local Conv1d features ($K=5$, 32 channels) are pooled by the same BiGRU stack before fusion with audio.

Dual signals encoding. E4 wristband and NeuroSky EEG are recorded on different hardware with different noise profiles and channel semantics. The **dual signals** ablation gives each sensor family its own encoder instead of one shared BiGRU over all six channels. E4 (EDA, HR, IBI) and EEG (theta, alpha, beta) each pass through a dedicated BiGRU with hidden size 64. Forward and backward states are concatenated per stream, giving a **128 dimensional** embedding for E4 and a **128 dimensional** embedding for EEG.

The default merge in this ablation is **concatenation** along the feature axis:

$$\mathbf{b} = [\mathbf{b}_{E4}; \mathbf{b}_{EEG}] \in \mathbb{R}^{256},$$

followed by the usual audio biosignal cross attention on $\mathbf{b}$. We also tested alternative **intra biosignal** merges, including cross attention with E4 as query and EEG as key/value (and the reverse), before the main fusion step. All of these dual signals variants scored below the single tower BiGRU baseline: separate encoders added parameters without better generalization under LOSO.

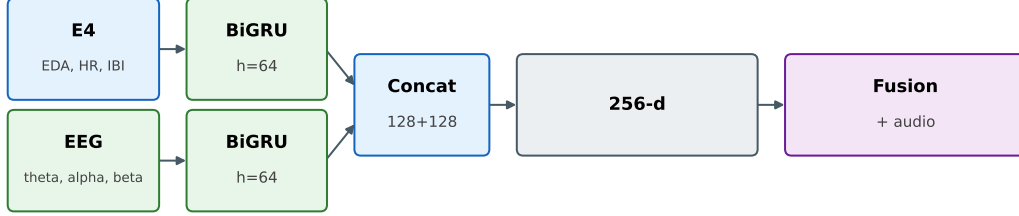


Figure 3: Dual-signals ablation: separate BiGRU towers for E4 and EEG ($h=64$ each) are concatenated to 256-d before cross-attention with audio.

Inter window temporal context (GRU/LSTM $\times 5$). Separate from within window encoding, we tested an **inter window** recurrent layer on top of fused embeddings. After cross attention produces a 256 dimensional vector per retained window, a unidirectional GRU or LSTM (hidden size 128) reads a **causal** sequence of the last five fused vectors for the same participant. Only the final time step is passed to the classification head.

The five vectors are ordered by **debate time** (window start seconds), using only windows that survived VAD and preprocessing. They are **not guaranteed to be contiguous on the five second grid**: if VAD rejected a slot, that timestamp is missing from the sequence and the context jumps over the gap while staying in chronological order. At the start of a debate, fewer than five prior windows exist and the sequence is left padded by repeating the earliest available window. This ablation asks whether overload builds gradually over roughly 25 seconds of accepted speech; it did not improve Macro-F1 over single window prediction.

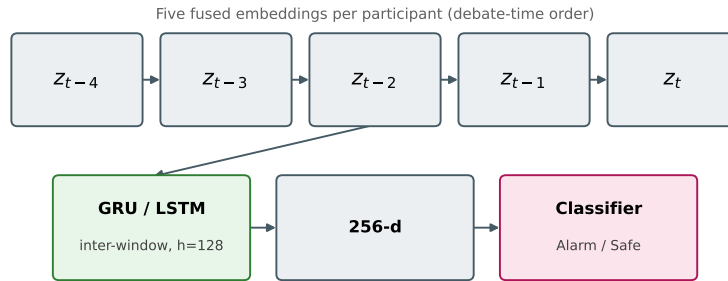


Figure 4: Inter-window ablation: a causal GRU or LSTM ($h=128$) reads the last five fused embeddings $z_{t-4}, \dots, z_t$ per participant; per-window encoders and cross-attention are unchanged.

Hand crafted feature MLP. HR and IBI are often sparse or missing for stretches of a five second window, so a sequence model may spend capacity on padding noise rather than stable structure. As an alternative biosignal path we extract **15 hand crafted scalars** from the raw (pre z score) six channel window during preprocessing:

- EDA: mean, standard deviation, linear slope, range (4 features);
- HR: mean, standard deviation, and RMSSD (Root Mean Square of Successive Differences, a standard metric for heart rate variability) (3);
- IBI: mean, standard deviation, and RMSSD on valid beats (3);
- EEG: mean θ , α , and β band power (3);
- EEG ratios: $\beta/(\theta + \alpha)$ and α/β (2). These ratios are established biomarkers for arousal and cognitive workload.

These summaries are z scored per participant across that participant’s windows, then passed through a small two layer MLP ($15 \rightarrow 128 \rightarrow 256$, ReLU, dropout) instead of the BiGRU. The raw 50×6 tensor is still stored but the encoder ignores temporal structure and reads only the feature vector. The rest of the model stays pooled cross attention with the fair training protocol. This variant targets interpretable HRV and EDA statistics rather than learned temporal convolutions.



Figure 5: Hand-crafted feature MLP: fifteen per-window scalars (EDA, HRV, EEG summaries) replace the sequence encoder before fusion with audio.

5.3 Pooled cross attention (main early fusion)

Both audio and biosignal embeddings are projected to 256 dimensions before fusion. Audio is the **query** and the pooled biosignal vector is a single **key/value** token (sequence length 1). Four head multi head attention produces an attended output $\mathbf{o}$. With one biosignal token the attention weight is always 1, so the attention step itself is identical across token modes; the models differ in how $\mathbf{o}$ is combined with the projected audio and biosignal embeddings afterward.

Our **main early fusion system** uses the standard audio anchored residual:

$$\mathbf{f} = \text{LayerNorm}(\mathbf{o} + \mathbf{q}_a),$$

where $\mathbf{q}_a$ is the projected audio query. No balanced residuals and no modality dropout. The biosignal still flows through attention into $\mathbf{o}$, but the skip path favors the pretrained audio representation. This is our best single network result (59.5% Macro-F1, Section 6).

All reported early fusion runs share the same fair protocol: weighted loss, balanced sampling, Macro-F1 checkpoint selection, frozen Wav2Vec 2.0, full preprocessing, and no augmentation unless an experiment states otherwise.

5.4 Other early fusion variants

We tried several alternatives to pooled cross attention on the same training protocol. None beat the pooled design on Macro-F1 or Alarm recall; they are summarized here for completeness and reported in Section 6.2.

Balanced cross attention (initial attempt). Our first fusion design tried to reduce audio dominance in the skip path. Instead of anchoring the residual on audio alone, we added **learnable scalars** w_a, w_b (initialized at 0.5) on both modalities:

$$\mathbf{f} = \text{LayerNorm}(\mathbf{o} + w_a \mathbf{q}_a + w_b \mathbf{k}_b),$$

where $\mathbf{k}_b$ is the projected biosignal token.

We also applied **modality dropout** during training (probability 0.15): each sample independently zeroed either the audio embedding or the biosignal embedding before fusion, but never both on the same sample. The goal was to force both branches to remain useful when one modality was missing.

On 27 participants this design **underperformed**: Alarm recall dropped sharply compared with the pooled design (Section 6). The extra regularization and dual residuals appear to have weakened the stable audio anchor that Wav2Vec 2.0 provides on this small dataset.

Quality aware cross attention. Quality tables and VAD metadata can inform fusion as well as augmentation. For each window we build a two dimensional **quality vector** $[q_{\text{audio}}, q_{\text{bio}}]$:

- q_{audio} : fraction of the five second span covered by speech (from VAD overlap);
- q_{bio} : mean sensor completeness for that participant (E4 and EEG tables from Section 3), in $[0, 1]$.

Before attention, the projected audio query and biosignal key are **scaled** by their quality scores (with a small floor so a modality is never fully zeroed). Standard pooled cross attention then runs on the scaled tokens. After attention, a learned **audio trust gate** $w \in (0, 1)$ combines the audio query with the attended biosignal output:

$$\mathbf{f} = \text{LayerNorm}(w\mathbf{q}_a + (1 - w)(\mathbf{o} + \mathbf{k}_b)),$$

where w is predicted from the concatenation of both embeddings and the quality vector. During training we add a small auxiliary loss (weight 0.1) that encourages w to match the normalized audio share $q_{\text{audio}}/(q_{\text{audio}} + q_{\text{bio}})$, so low speech overlap or poor biosignal completeness can down weight the audio skip path in a data driven way.

This design differs from **quality weighted late fusion** (Section 5.5), which combines already trained unimodal probabilities at decision time rather than modulating embeddings inside the network.

Sequence cross attention. In **sequence** mode the BiGRU returns all 50 time step outputs instead of a single pooled vector. Audio remains one query token; biosignals supply 50 key/value tokens. Attention can distribute weight over biosignal time steps and weights are stored for explainability plots. Macro-F1 is slightly below pooled cross attention in our experiments.

Gated multimodal unit (GMU). The **Gated Multimodal Unit** [5] learns a per dimension gate between audio and biosignal embeddings:

$$\mathbf{h}_a = \tanh(W_a \mathbf{a}), \quad \mathbf{h}_b = \tanh(W_b \mathbf{b}), \quad \mathbf{z} = \sigma(W_z [\mathbf{a}; \mathbf{b}]), \quad \mathbf{h} = \mathbf{z} \odot \mathbf{h}_a + (1 - \mathbf{z}) \odot \mathbf{h}_b.$$

We add light gate regularization and a slight audio biased initialization so training starts stable when the audio branch is pretrained and the biosignal branch is random. GMU underperformed pooled cross attention on this dataset.

5.5 Late fusion at decision level

Early fusion models (Section 5.3) learn a shared representation inside one network. **Late fusion** takes a different path: each modality (or sensor family) is trained as a **separate classifier**, and predictions are combined only at **decision time** using the saved Alarm probabilities. No fusion trunk is retrained. This lets each stream specialize and avoids forcing E4, EEG, and audio into one biosignal encoder when they behave differently.

Base classifiers. Every late fusion model reuses the same overall architecture (Wav2Vec 2.0 + BiGRU + cross attention + linear head), but is trained with only one modality active:

- **Audio only:** the biosignal branch is zeroed after encoding.
- **Bio only (2 mod setup):** audio is zeroed; one BiGRU reads all six physio channels.
- **E4 only / EEG only (3 mod setup):** audio is zeroed; each stream uses dual signals encoding on its three channels without the other sensor group.

Each base model is trained with the same LOSO protocol, weighted loss, balanced sampling, and Macro-F1 checkpoint selection as early fusion (Section 5.6). For each held out participant we load the corresponding fold checkpoints from every base model and predict on the same windows.

Fusion protocol. Within each LOSO fold, base models first predict on a **validation** subset of the training participants (one held out validation subject, using the same validation split as during model training). Methods that need fitted weights (validation F1 weighting, stacking) use only these validation predictions and labels. The held out test participant is never seen during fusion weight fitting. Test predictions from all base models are aligned window by window before fusion. Let $p_m(i)$ denote the predicted Alarm probability from modality $m \in \{1, \dots, M\}$ for window i , with $M = 2$ (audio + bio) or $M = 3$ (audio + E4 + EEG).

Uniform average. The simplest rule averages Alarm scores with equal weight:

$$\hat{p}(i) = \frac{1}{M} \sum_{m=1}^M p_m(i).$$

Windows are labeled Alarm if $\hat{p}(i) \geq 0.5$. This baseline treats every stream as equally reliable and does not adapt to class imbalance or per modality calibration.

Majority vote (logical OR). A conservative **any stream alarms** rule fires Alarm when *any* base model exceeds threshold:

$$\hat{y}(i) = \mathbb{I}[\exists m : p_m(i) \geq 0.5].$$

The fused score is 0 or 1 (no soft probability). OR fusion boosts Alarm recall because a single sensitive modality can trigger detection, at the cost of more false alarms than stacking in some runs.

Quality weighted average. Quality weighted fusion scales each stream by a **per window** reliability score, then normalizes:

$$\hat{p}(i) = \frac{\sum_{m=1}^M w_m(i)}{\sum_{m'} w_{m'}(i)} p_m(i).$$

For audio, $w_{\text{audio}}(i)$ is the fraction of the five second window covered by speech (from VAD overlap metadata). For E4 and EEG streams, w is the participant level sensor completeness score from the K EmoCon quality tables (Section 3). Low quality windows down weight the corresponding modality. In our experiments this rule was **too conservative**: it under predicted Alarm compared with stacking or OR.

Validation F1 weighted average. An adaptive variant assigns **fixed weights per fold** from each base model’s Alarm F1 on the validation split:

$$w_m = \frac{\text{F1}_m^{\text{val}}}{\sum_{m'} \text{F1}_{m'}^{\text{val}}}, \quad \hat{p}(i) = \sum_{m=1}^M w_m p_m(i).$$

Streams that discriminate Alarm better on validation receive higher weight. Unlike quality weighting, this uses **task performance**, not sensor metadata.

Stacking logistic regression. The strongest late fusion method fits a **logistic regression meta learner** on validation Alarm probabilities. For each validation window the feature vector is $\mathbf{x} = [p_1, \dots, p_M]$. A balanced class weight logistic regressor learns $\hat{p} = \sigma(\mathbf{w}^\top \mathbf{x} + b)$ on validation labels, then outputs soft Alarm scores on the test fold. Stacking can up weight a modality when its probabilities are calibrated and informative, and down weight redundant or noisy streams. It requires both classes in validation; otherwise the implementation falls back to uniform averaging.

Two and three modality setups are evaluated:

- **Two modalities:** audio + single biosignal model (all six physio channels).
- **Three modalities:** audio + E4 only + EEG only, each trained independently.

Stacking LR with three modalities achieved the best overall Macro-F1 in this project (Section 6).

Stacking with per participant physio normalization. Our main preprocessing (Section 4.2) z scores each channel **inside each five second window**. That stabilizes scale for the BiGRU but removes slow baseline shifts across the debate (for example a rising EDA level over several minutes). To test whether keeping those tonic levels helps biosignals, we ran a **preprocessing variant** with per participant normalization: statistics are computed once over all retained windows of a debater and applied to every window from that person.

Under this variant we **rebuilt** labels, audio, physio, and window tensors, then retrained **audio only** and **bio only** base classifiers from scratch with the same fair training settings. We applied **stacking logistic regression** on the new probability outputs (two modalities: audio + bio). Audio preprocessing is unchanged; only biosignal scaling differs. The question is whether late fusion gains when the bio stream can read absolute within participant levels instead of window relative fluctuations.

5.6 Training objective and sampling

Classification head. Every early fusion variant (pooled, sequence, GMU, and unimodal runs used in late fusion) ends in the same **binary classification head**. The fused representation $\mathbf{f} \in \mathbb{R}^{256}$ is passed through one fully connected layer with bias:

$$\mathbf{z} = W\mathbf{f} + \mathbf{b}, \quad \mathbf{z} \in \mathbb{R}^2,$$

where z_0 and z_1 are raw logits for Safe and Alarm respectively. There is no extra hidden layer, dropout, or nonlinearity on the head: all nonlinear processing happens upstream in the encoders and fusion block.

During training, logits are fed to **weighted cross entropy** (below). The loss applies a softmax internally and up weights Alarm windows through class weights $w_{\text{Safe}}, w_{\text{Alarm}}$. Gradients therefore adjust the single weight matrix W and bias $\mathbf{b}$ to separate the two classes in the 256 dimensional fusion space.

At validation and test time we apply a softmax to obtain class probabilities $p_c = e^{z_c} / \sum_{c'} e^{z_{c'}}$. The **hard prediction** is $\hat{y} = \arg \max_c z_c$, which for two classes is equivalent to labeling Alarm when $p(\text{Alarm}) > 0.5$. We store both the discrete labels and the Alarm probability $p(\text{Alarm})$ per window. Macro-F1 checkpoint selection and early stopping use the hard predictions on the validation split; saved Alarm probabilities are reused later for late fusion and threshold analysis (Section 6).

When the inter window temporal model is enabled (Section 5.2), the head receives the final temporal state instead of a single window fusion vector, but its form stays a single linear map to two logits.

Class weighted loss. The class distribution in Section 2 is heavily Safe biased. Without countermeasures the model learns to predict Safe almost always. Training minimizes weighted cross entropy over the two classes. For class $c \in \{\text{Safe}, \text{Alarm}\}$ let N_c be the number of training windows with label c in the current fold and $N_{\text{total}} = N_{\text{Safe}} + N_{\text{Alarm}}$. Class weights are computed using the standard balanced heuristic:

$$w_c = \frac{N_{\text{total}}}{2 N_c}. \quad (1)$$

Because $N_{\text{Alarm}} \ll N_{\text{Safe}}$, the Alarm class receives a proportionally larger weight. This specific formulation ensures that the expected sum of weights for each class over the entire dataset is equal ($N_c \cdot w_c = N_{\text{total}}/2$). By equating the total loss contribution of the rare Alarm class to that of the majority Safe class, we prevent the Safe majority from dominating the gradient updates. The final weighted loss is $\mathcal{L} = -\sum_i w_{y_i} \log p(y_i | \mathbf{x}_i)$.

Balanced sampling across participants. Some participants contribute many more windows than others after VAD filtering. Each training epoch samples exactly

$$K = \text{median}(n_i)$$

windows per participant i , where n_i is that participant’s window count in the training fold. Indices are drawn at random with replacement when needed. This stops a few talkative subjects from dominating gradient updates.

Other training settings used in reported experiments. Validation uses the same LOSO folds as Section 3. The checkpoint with best validation **Macro-F1** is kept. Typical settings are batch size 8, early stopping patience 8, and at least 5 epochs before stopping can trigger. Augmentation is off unless an experiment explicitly uses the offline augmented tensors (Section 4.3).

6 Results

All tables in this section use the **synthetic 50/50** metric protocol from Section 2 unless noted otherwise. Rows list the five report metrics. **Bold** in the Macro-F1 row marks the best value in each table group. Experiments follow Leave One Subject Out validation on 27 participants with weighted loss, balanced sampling, Macro-F1 checkpoint selection, frozen Wav2Vec 2.0, and no augmentation unless stated.

6.1 Early fusion: pooled cross attention

The **best early fusion model** is pooled cross attention with weighted loss and balanced sampling (59.50% Macro-F1 under synthetic 50/50, 51.2% Alarm recall). This is the end to end network we designed and ablated throughout the report. The overlap analysis and ROC panels below refer to this system.

	Synthetic 50/50	Real class split	Δ (pp)
Accuracy	59.80	65.68	+5.88
Recall (Alarm)	51.20	51.25	+0.05
F1 (Alarm)	56.02	32.45	−23.57
Precision (Alarm)	61.84	23.75	−38.09
Macro-F1	59.50	54.73	−4.77

Table 3: Best pooled cross attention: synthetic 50/50 versus real class distribution ($n = 1492$, 83.9% Safe). Δ is real minus synthetic in percentage points.

On the real 84/16 split, Accuracy rises (+5.9 pp) but Macro-F1 falls (−4.8 pp) because Safe dominates. Alarm recall is unchanged (+0.05 pp); precision and F1 drop sharply. Synthetic 50/50 metrics (Section 2) are used for fair comparison across models. The pooled architecture and training setup are in Sections 5.3 and 5.6.

Figure 6 shows the normalized confusion matrix for the best pooled model.

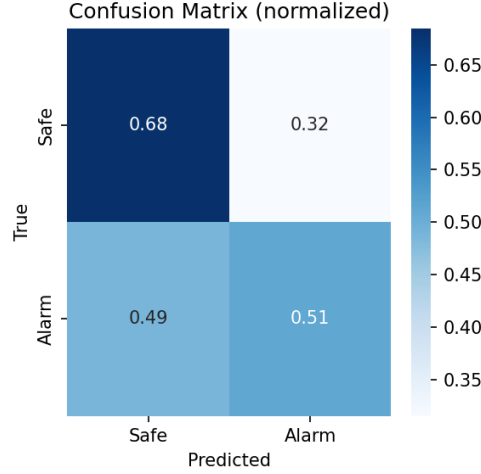


Figure 6: Normalized confusion matrix for the best pooled cross attention model (LOSO pooled test predictions).

Modality overlap analysis. To understand early fusion behavior we compared window level predictions from audio only, bio only, and pooled cross attention fusion on the same 1,492 windows (pooled weighted model). Unimodal accuracy on the real label split is 75.0% (audio) and 73.3% (bio), but fusion accuracy is only 65.7%. An **oracle** that picks the correct modality whenever at least one stream is right would reach 86.0% accuracy, so fusion leaves headroom.

When audio and bio disagree (23.7% of windows), fusion follows the **audio prediction 80.5%** of the time. In 164 windows bio is correct and audio is wrong; fusion copies the wrong audio label in 131 cases (79.9%). This matches the audio anchored residual in pooled cross attention (Section 5.3) and motivates late fusion: independently trained streams plus stacking can recover bio only cues that joint early fusion suppresses.

There is still synergy: in 84 windows (5.6%) both unimodal models are wrong but fusion is right. Interference also occurs: 220 windows (14.8%) where both unimodals are correct but fusion fails. Overall, early fusion trades some bio information for audio stability; late fusion stacking partially reverses that trade off at decision level.

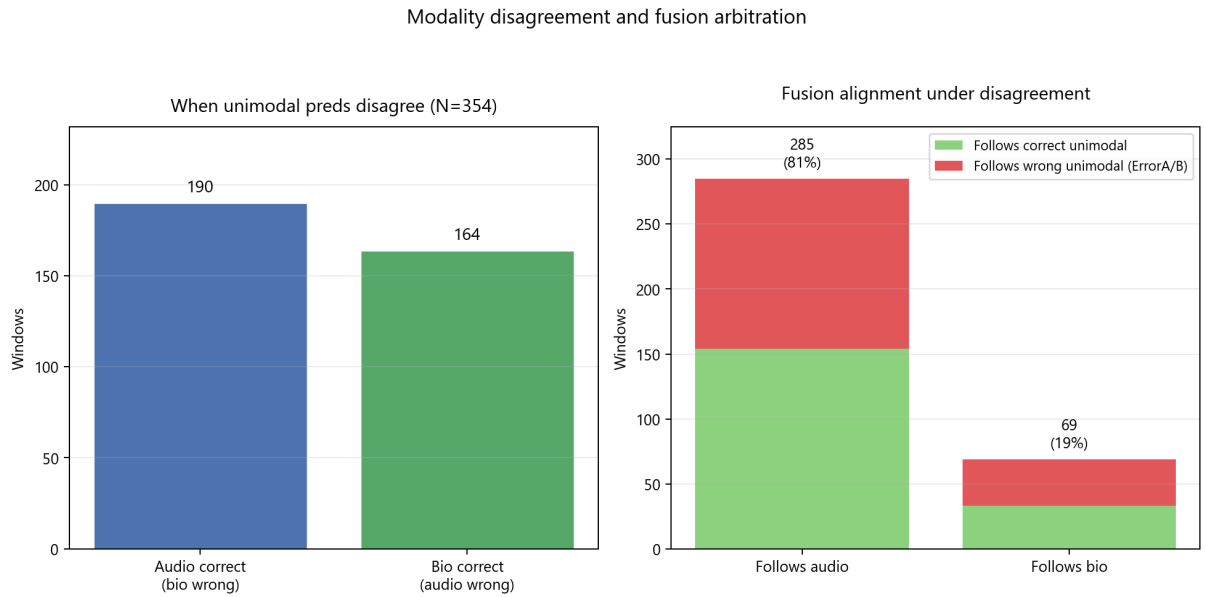


Figure 7: Modality disagreement and fusion arbitration on pooled LOSO test windows ($n = 1492$). When unimodals disagree, fusion follows audio in 80.5% of cases.

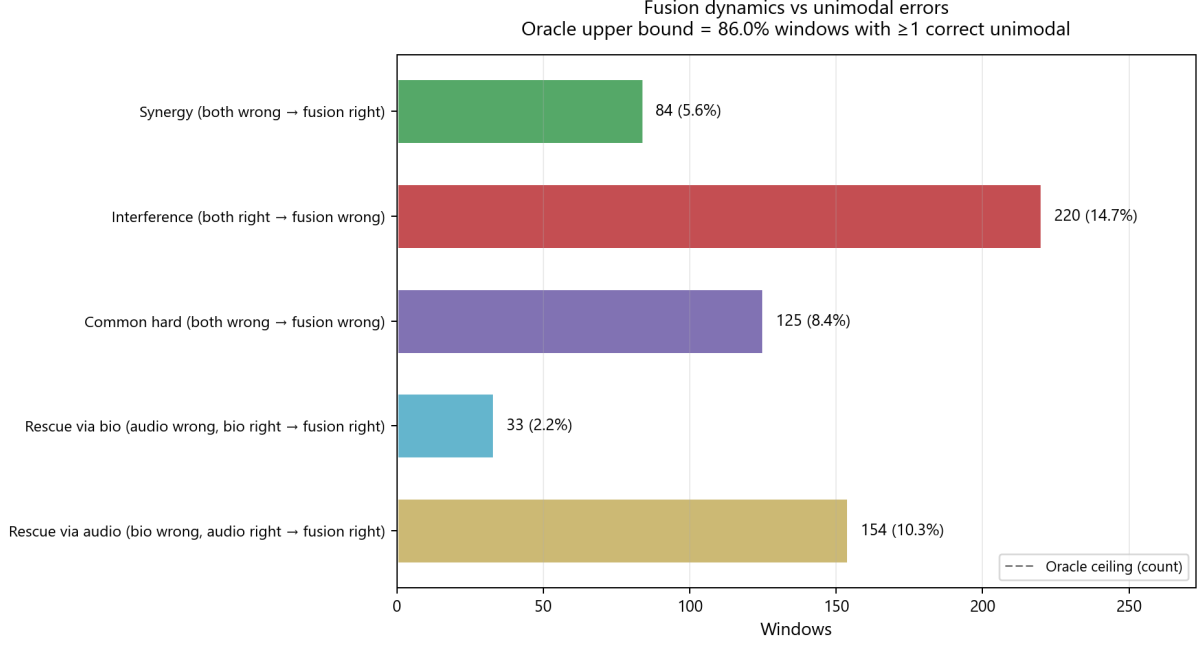


Figure 8: Fusion dynamics relative to unimodal errors. Interference (220 windows, 14.7%) exceeds synergy (84 windows, 5.6%). The dashed line marks the oracle count where at least one unimodal model is correct.

Interference is the largest bar: both unimodals are correct but fusion fails on 220 windows (14.7%). Rescue via audio (154 windows) is larger than rescue via bio (33 windows). The oracle ceiling at 86.0% shows how much room a better arbitration rule could still recover.

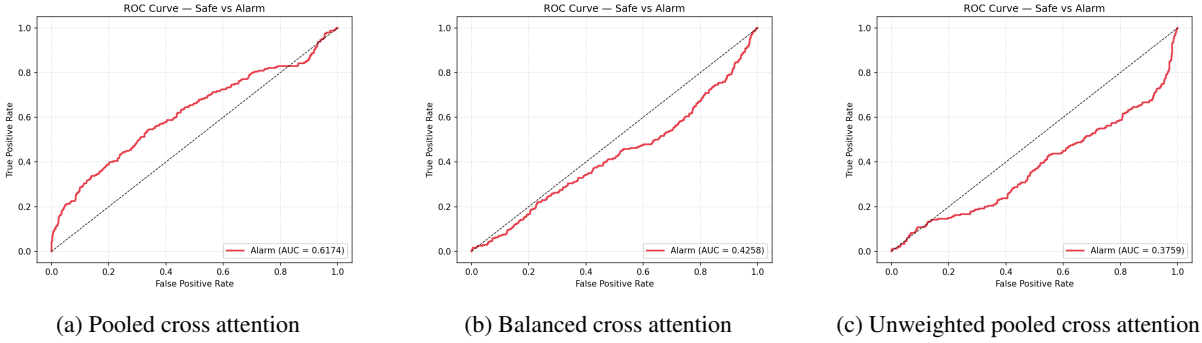


Figure 9: Early fusion ROC curves on pooled LOSO test predictions (Alarm class). Panel (a) is the best early fusion model.

Panel (a) is pooled cross attention with an audio anchored residual (Section 5.3) and class weighted loss plus balanced per participant sampling (Section 5.6). The curve rises away from the diagonal, matching 51.2% Alarm recall in Table 4. Panel (b) uses the balanced fusion design (Section 5.4); the curve hugs the diagonal (12.0% Alarm recall). Panel (c) turns off weighted loss and balanced sampling (Section 5.6); the curve is nearly flat (1.6% Alarm recall).

6.2 Early fusion alternatives and ablations

Table 4 and Table 5 collect every alternative early fusion run and component swap against the same pooled reference. Architectures are defined in Sections 5.4, 5.2, and 4.3. Interpretation paragraphs follow each table.

Metric	Ref.	Fusion architecture changed				Training changed	
	Pooled	Balanced	Sequence	GMU	Quality	Unweighted	Aug.
Accuracy	59.80	48.30	57.00	52.00	49.70	49.80	46.90
Recall (Alarm)	51.20	12.00	43.40	21.20	11.60	1.60	11.60
F1 (Alarm)	56.02	18.84	50.23	30.64	18.74	3.09	17.93
Precision (Alarm)	61.84	43.80	59.62	55.21	48.74	44.44	39.46
Macro-F1	59.50	40.45	56.19	46.97	41.16	34.61	39.34

Table 4: Early fusion and training variants (synthetic 50/50). **Pooled** = reference (pooled cross attention, weighted loss, balanced sampling, no augmentation). Other columns change only the component named in the header.

Fusion architecture. GMU, balanced residuals with modality dropout, sequence cross attention, and quality aware cross attention all trail pooled fusion on Macro-F1 and Alarm recall (Section 5.4). The extra gates and regularization weaken the stable Wav2Vec 2.0 audio anchor on 27 participants. Scaling fusion inputs by speech overlap and sensor completeness (Section 5.4) underperforms on all five metrics; the auxiliary trust gate appears to over correct toward Safe. This is distinct from quality gated augmentation (Section 4.3) and quality weighted late fusion (Table 6).

Training. Removing class imbalance handling (unweighted column) collapses Alarm recall to 1.6%. Offline augmentation **hurts** every Alarm metric (Macro-F1 -20.2 pp, Alarm recall -39.6 pp). Quality gated sensor noise and SpecAugment style audio perturbations add variance without helping a 27 participant corpus.

Metric	Ref.	Component swapped (vs pooled)				
	Pooled	CNN	Feat. MLP	Dual-sig.	GRU $\times 5$	LSTM $\times 5$
Accuracy	59.80	51.10	54.50	52.10	56.70	51.10
Recall (Alarm)	51.20	20.80	23.40	24.20	29.20	18.40
F1 (Alarm)	56.02	29.84	33.96	33.56	40.28	27.34
Precision (Alarm)	61.84	52.79	61.90	54.75	64.89	53.18
Macro-F1	59.50	46.16	49.63	48.06	53.16	45.25

Table 5: Component ablations on the pooled early fusion stack (synthetic 50/50). Each column replaces one module; all other settings match the pooled reference.

Biosignal encoder. The default BiGRU encoder (Section 5.1) wins on every metric. The 1D CNN front end and hand crafted feature MLP sacrifice Alarm recall (20.8% and 23.4% vs 51.2%), consistent with overfitting or lost temporal structure on 27 participants. Separate E4 and EEG encoders with concatenation (Section 5.2) reach only 48.06% Macro-F1 and 24.2% Alarm recall; extra parameters do not generalize better under LOSO.

Inter window temporal context. Causal GRU and LSTM over the last five fused vectors (Section 5.2) both lag single window pooled fusion. Alarm recall drops by 22.0 pp (GRU $\times 5$) and 32.8 pp (LSTM $\times 5$). Overload cues appear mainly inside each five second segment; retained windows are not contiguous (Section 4.1), so inter window context adds noise.

6.3 Late fusion

Table 6 lists all decision level fusion rules for two modality (audio + bio) and three modality (audio + E4 + EEG) setups. Base classifiers and rules are in Section 5.5.

Metric	Quality		Stacking LR		Majority OR		Uniform		Val-F1	
	2m	3m	2m	3m	2m	3m	2m	3m	2m	3m
Accuracy	51.90	52.30	59.30	62.30	59.00	59.60	50.90	51.40	49.80	49.30
Recall (Alarm)	18.40	14.20	47.60	51.60	48.40	48.40	15.40	10.00	10.00	8.00
F1 (Alarm)	27.67	22.94	53.91	57.78	54.14	54.50	23.88	17.06	16.61	13.63
Precision (Alarm)	55.76	59.66	62.14	65.65	61.42	62.37	53.10	58.14	49.02	45.98
Macro-F1	45.82	44.20	58.74	61.86	58.53	59.09	43.82	41.35	40.35	38.87

Table 6: Late fusion decision rules (synthetic 50/50). **2m** = audio + bio; **3m** = audio + E4 + EEG. Stacking LR with three modalities is the best overall system (61.86% Macro-F1).

Stacking logistic regression and majority vote (OR) are the only rules that recover usable Alarm recall (~ 48 – 52%). Quality weighting, uniform averaging, and validation F1 weighting stay below 19% recall because down weighting

weak streams suppresses rare Alarm events. Adding a third modality helps stacking (+3.1 pp Macro-F1, +4.0 pp Alarm recall) but not quality weighting. OR is stable across two and three streams.

Stacking LR with three modalities splits E4 and EEG into separate base classifiers so the meta learner can combine complementary errors. A separate run with per participant physio normalization and two modality stacking (audio + bio) reaches 53.89% Macro-F1 and 34.6% Alarm recall (Table 6.3); that comparison also differs in modality count (Section 4.2).

6.4 Best late fusion

The **best late fusion system** is stacking logistic regression over three independently trained classifiers (audio, E4, EEG). It achieves the highest Macro-F1 in the project (61.86% under synthetic 50/50, Table 6). Section 5.5 describes the decision rules and base models.

	Synthetic 50/50	Real class split	Δ (pp)
Accuracy	62.30	69.50	+7.20
Recall (Alarm)	51.60	51.67	+0.07
F1 (Alarm)	57.78	35.28	-22.51
Precision (Alarm)	65.65	26.78	-38.87
Macro-F1	61.86	57.66	-4.20

Table 7: Best late fusion (stacking LR, 3 modalities): synthetic 50/50 versus real class distribution ($n = 1492$, 83.9% Safe).

The pattern matches early fusion: Alarm **recall is stable** across splits (+0.07 pp) while Accuracy rises on the imbalanced real distribution (+7.2 pp) and Alarm precision/F1 fall sharply. Macro-F1 drops by 4.2 pp on the real split, less than for pooled cross attention (-4.8 pp).

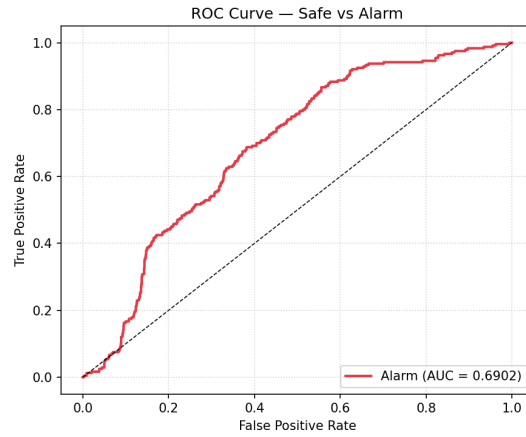


Figure 10: ROC curve for the best late fusion system (stacking LR, 3 modalities) on pooled LOSO test predictions.

The late fusion ROC bends further toward high Alarm sensitivity than the pooled early fusion curve in panel (a) of Figure 9, consistent with the higher synthetic Macro-F1 (61.86% vs 59.50%).

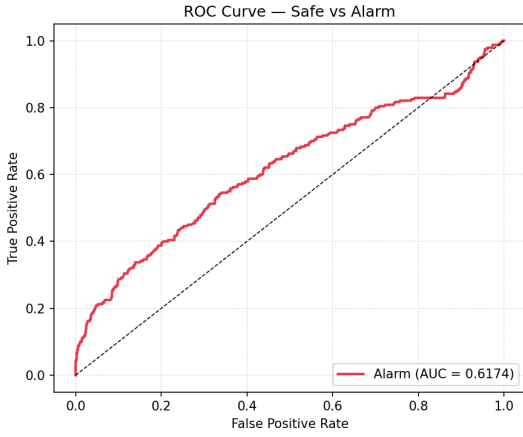
6.5 Best overall comparison

Table 8 compares the two best systems on the five report metrics under synthetic 50/50 and on the real 84/16 class distribution from the same LOSO predictions.

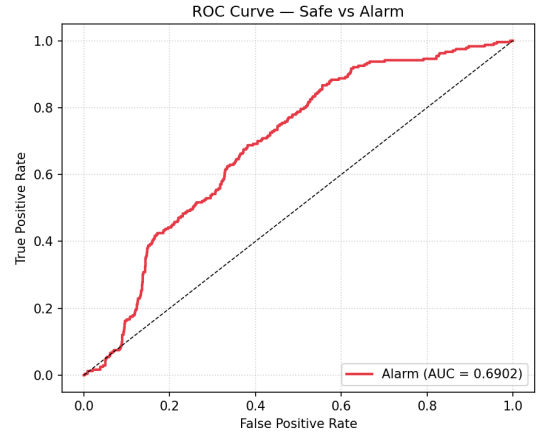
	Synthetic 50/50		Real class split	
	Early fusion	Late fusion 3m	Early fusion	Late fusion 3m
Accuracy	59.80	62.30	65.68	69.50
Recall (Alarm)	51.20	51.60	51.25	51.67
F1 (Alarm)	56.02	57.78	32.45	35.28
Precision (Alarm)	61.84	65.65	23.75	26.78
Macro-F1	59.50	61.86	54.73	57.66

Table 8: Best early fusion (pooled cross attention) versus best late fusion (stacking LR, 3 modalities).

Under **synthetic 50/50**, late fusion wins on every metric, with Macro-F1 +2.4 pp and Alarm recall +0.4 pp. Under the **real imbalanced split**, both systems keep roughly the same Alarm recall ($\sim 51.3\%$), so class skew does not change which stream detects overload windows. Late fusion still leads on Macro-F1 (+3.0 pp) and Alarm F1 (+2.8 pp) on the real split, while raw Accuracy is misleading for both (Safe is 84% of windows). For deployment on this label distribution, **late fusion stacking with three modalities** is the stronger choice; pooled cross attention remains the best **single trained network** and the reference for ablations and overlap analysis.



(a) Best early fusion (pooled cross attention)



(b) Best late fusion (stacking LR, 3 modalities)

Figure 11: ROC comparison of the two best systems on pooled LOSO test predictions (Alarm class).

Side by side ROC curves on the **natural class distribution** show that late fusion ranks Alarm windows slightly better across thresholds, matching Table 8. This comparison is more representative of deployment than synthetic 50/50 tables alone, because both curves use the same imbalanced test labels. Synthetic 50/50 remains useful for fair ranking across experiments (Section 2); the dual ROC figure shows how that ranking translates to real prevalence.

7 Conclusion

This report presented Brain Drain Detector, a multimodal system that classifies five second debate windows as Safe or Alarm from self reported arousal and valence. After preprocessing, evaluation used 1,492 windows from 27 participants under strict Leave One Subject Out validation and a synthetic 50/50 metric protocol for fair comparison. The **best overall system** was **late fusion stacking with three modalities** (61.86% Macro-F1, 51.6% Alarm recall under synthetic 50/50). The best **single end to end network** was **pooled cross attention** with weighted training (59.50% Macro-F1, 51.2% Alarm recall). Side by side ROC curves on the real class distribution (Figure 11) confirm that late fusion ranks Alarm windows slightly better at deployment prevalence (Section 6.5).

7.1 Effective components

Class imbalance handling. Weighted cross entropy and balanced per participant sampling (Section 5.6) were necessary. Without them, Alarm recall fell to 1.6% and ROC curves collapsed toward the majority Safe class (Table 4, Figure 9). This matches the imbalance reported for K EmoCon, where Likert scale annotations are heavily biased toward neutral states [1].

Simple early fusion. Pooled cross attention with an audio anchored residual (Section 5.3) outperformed the initial balanced design with dual residuals and modality dropout, sequence cross attention, and GMU [5]. On 27 held out subjects, a lighter fusion path with a stable Wav2Vec 2.0 anchor generalized better than more flexible gates.

Late fusion at decision time. Training audio, E4, and EEG as separate classifiers and combining probabilities with stacking logistic regression achieved the best overall Macro-F1 (61.86%, Table 6, Section 6.5). The modality overlap analysis (Section 6.1) helps explain why: early fusion often follows audio when streams disagree, while stacking can up weight a stream when its probabilities are informative (Section 5.5).

Simple biosignal encoding. A single tower BiGRU over six channels beat a 1D CNN front end, dual E4/EEG towers, hand crafted feature MLP, and inter window GRU/LSTM stacks (Table 5). Overload cues in our setup appear to live mainly inside each retained five second segment, not across sparse VAD gaps.

7.2 Underperforming approaches

Richer fusion and encoder variants. Balanced cross attention, quality aware fusion, GMU, dual signals encoding, and inter window temporal models all underperformed the pooled BiGRU baseline on Macro-F1 and Alarm recall. These designs add parameters or inductive biases that need more data or cleaner signals to pay off.

Offline augmentation. Quality gated sensor noise and SpecAugment style audio perturbations hurt every Alarm metric (Table 4). On a 27 participant corpus, synthetic perturbations appear to add variance without enlarging the effective training set.

Quality and normalization shortcuts. Quality weighted late fusion, quality scaling inside the fusion layer, and per participant physio normalization did not improve on the main pipeline (Tables 6, 4, 6.3). Down weighting weak streams or tonic baselines tended to push predictions toward Safe.

7.3 Dataset limitations

K EmoCon is valuable because it records multimodal signals during naturalistic conversation [1], but it is a **small and difficult** corpus for our binary overload task. The original release contains 16 debate sessions and 32 participants; after speaker separation and speech overlap filtering we retain 1,492 windows with roughly **83% Safe and 17% Alarm**. Six participants in the public release have incomplete E4 recordings due to device or collection issues, which is noted in follow up work on the same corpus [2].

The dataset authors report several limitations that align with our experience. Contact EEG is sensitive to movement and facial muscle activity; debate settings may encourage participants to regulate emotional expression; and the participant pool is young, highly educated, and not demographically diverse [1]. Annotations on arousal and valence are skewed toward neutral, with only a small fraction of labels in extreme states [1]. Under Leave One Subject Out, each test fold sees a new speaker with little training data for that identity, which is stricter than random window splits used in some published K EmoCon studies [2].

Our absolute Macro-F1 values (roughly 42% to 62% under synthetic 50/50) should be read in that context. They show that overload related patterns are partially detectable, but not that the problem is solved for deployment.

7.4 Generalization and alternative methods

Gonçalves *et al.* [2] show that a heavier convolution augmented transformer can reach strong Macro-F1 on K EmoCon for arousal and valence classification from physiological signals alone, using focal loss for class imbalance and a different train test protocol (random 80/20 splits with five fold cross validation, not LOSO). That result does not transfer directly to our binary speech plus biosignal task, but it shows that the corpus can support more complex models under some conditions.

A larger dataset with more Alarm windows per participant, cleaner wearable sensing, and broader demographics would likely help methods that failed in our study. Dual stream encoders, inter window context, quality aware fusion, and augmentation are standard tools in multimodal affective computing; they probably need more labeled data and more stable physiology to generalize under subject independent evaluation. Park *et al.* [1] argue that debate formality and sensor noise limit how well current K EmoCon data represent emotions in the wild, so collecting new data in less constrained settings remains important.

7.5 Summary and future work

For Brain Drain Detector on K EmoCon, the best overall recipe is late fusion stacking over audio, E4, and EEG specialists. The best single network recipe is frozen Wav2Vec 2.0, a simple BiGRU biosignal encoder, and pooled cross attention with explicit class imbalance handling. Complexity should be added only when the data support

it. Future work should prioritize more participants, more overload positive windows, and subject independent evaluation before chasing larger fusion architectures on this corpus alone.

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